

MAY 2026

DATA ANALYTICS CASE STUDY

E-COMMERCE SALES ANALYSIS

USING MICROSOFT EXCEL & SQL

An analysis of sales performance, customer behavior,
and regional trends to derive actionable business
insights and recommendations.

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E-Commerce Sales Data Analysis

Business Intelligence Case Study Report

Executive Summary

This report presents the findings of a comprehensive data analytics project conducted on the Superstore Sales Dataset, a publicly available e-commerce dataset sourced from Kaggle.

The objective of this project was to analyze sales performance, customer behavior, regional trends, and product category performance using Excel and SQL. The project demonstrates the complete data analytics workflow including data cleaning, SQL analysis, dashboard development, and business recommendations.

The dataset contains approximately 9,800 records and 18 columns covering multiple business dimensions such as:

- Order Dates
- Shipping Modes
- Customer Segments
- Regions and States
- Product Categories
- Sales Revenue

The analysis involved:

- Cleaning and transforming data in Microsoft Excel
- Writing 30+ SQL queries for analysis
- Creating an interactive Excel dashboard
- Generating actionable business insights and recommendations

Key Findings

- Total sales exceeded approximately **\$2.3 Million**
- The Consumer Segment generated the highest revenue
- Technology was the top-performing category
- California and New York were the best-performing states
- Sales showed consistent year-over-year growth
- Q4 (October–December) produced peak sales performance

Using Microsoft Excel & SQL

Dataset: Superstore Sales Dataset (Kaggle)

Records: 9,800 Transactions

Tools Used: Microsoft Excel, SQL, GitHub

Project Type: Business Intelligence / Portfolio Project

Analyst: Siddique Malik

Date: May 2026

1. Introduction

In today's competitive business environment, organizations rely heavily on data-driven decision-making to improve performance and optimize business strategy.

This project simulates a real-world data analytics engagement using the standard data analysis workflow:

- Ask
- Prepare
- Process
- Analyze

- Share
- Act

The project applies Microsoft Excel and SQL to transform raw sales data into meaningful business insights.

The primary objective of this case study is to identify:

- High-performing regions
- Strong customer segments
- Revenue-driving product categories
- Sales trends over time
- Strategic growth opportunities

2. Business Problem (ASK Phase)

The primary business problem addressed in this project is:

“How can an e-commerce business use historical sales data to better understand revenue drivers, optimize regional performance, and improve customer engagement?”

Business Questions

1. Which regions and states generate the highest revenue?
2. Which product categories contribute the most sales?
3. Which customer segment performs best?
4. How do monthly and yearly sales trends behave?
5. Which cities generate the highest sales?
6. What shipping modes are most frequently used?

Stakeholders

- Business Management Team
 - Sales Department
 - Marketing Team
 - Regional Managers
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3. Dataset Information (PREPARE Phase)

Dataset Details

Dataset Name: Superstore Sales Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>

Dataset Overview

- Approximately 9,800 rows
- 18 columns
- Multi-year sales data
- CSV file format

Important Columns

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Customer Segment

- Region
 - State
 - City
 - Category
 - Sales
-

Data Credibility (ROCCC Framework)

Reliable

The dataset is widely used by analysts and students for analytics projects.

Original

The dataset represents realistic e-commerce transactional records.

Comprehensive

It includes geography, customer segments, categories, and sales information.

Current

The dataset spans multiple years allowing trend analysis.

Cited

The dataset is publicly available on Kaggle for educational use.

4. Data Cleaning & Processing (PROCESS Phase)

Data cleaning and preprocessing were performed in Microsoft Excel before conducting SQL analysis and dashboard development.

Cleaning Steps Performed

- Removed duplicate records
- Standardized date formats
- Organized and formatted columns
- Checked for missing values
- Validated numeric data types
- Standardized categorical values
- Exported clean data for SQL analysis

Additional Columns Created for Analysis

To improve reporting, pivot table analysis, and dashboard visualization, several calculated columns were added:

Added Column	Purpose
Year	Extract yearly sales trends
Month Name	Analyze monthly sales patterns
Month Number	Sort months correctly in charts
Quarter	Analyze quarterly business performance
Shipping Days	Measure delivery duration
Order Count	Track total transaction frequency

These additional fields improved data organization and enabled more advanced trend analysis and dashboard filtering.

Tools Used

- Microsoft Excel
- Pivot Tables

- Excel Formulas
- Filters & Sorting
- Remove Duplicates Tool
- Date Formatting Functions

After cleaning and transformation, the dataset was prepared for SQL analysis and dashboard creation

5. SQL Analysis (ANALYZE Phase)

More than 30 SQL queries were written and executed to analyze the dataset.

Analysis Performed

- Total sales analysis
- Sales by region
- Sales by category
- Top-performing cities and states
- Monthly sales trends
- Yearly growth analysis
- Customer segment analysis
- Shipping mode analysis

Sample SQL Query

```
SELECT
    Region,
    ROUND(SUM(Sales), 2) AS Total_Sales
FROM superstore_sales
```

GROUP BY Region

ORDER BY Total_Sales DESC;

Additional SQL Tasks

- Revenue aggregation
- Trend analysis
- Ranking top-performing locations
- Comparing customer segments
- Monthly and yearly grouping

6. Dashboard & Visualizations (SHARE Phase)

An interactive Excel dashboard was created to visualize business performance.



Dashboard Components

- KPI Cards
- Sales by Region
- Monthly Sales Trend
- Sales by Category
- Sales by Segment
- Top Cities by Sales
- Sales by State Map
- Yearly Performance Charts

Dashboard Features

- Interactive Filters
- Region Selection
- Year Filters
- Segment Filters
- Ship Mode Filters

The dashboard provides a high-level executive overview of sales performance.

7. Key Findings & Insights

1.Revenue Performance

- Total sales reached approximately **\$2.3 Million** across the dataset period.
- The business processed a total of **9,800 orders**.
- The average order value was approximately **\$231**
- Revenue performance indicates strong overall sales activity.
- Sales volume remained consistent across multiple years.

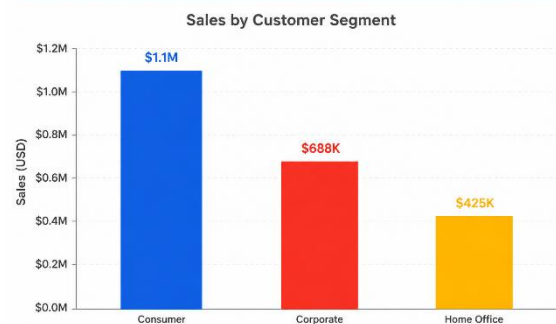
1. REVENUE PERFORMANCE



2.Customer Segment Analysis

- **Consumer** segment generated the **highest** revenue contribution at approximately **\$1.1M**.
- Corporate customers contributed nearly **\$688K** in total sales.
- Home Office segment generated around **\$425K** in revenue.
- Consumer customers remained the primary drivers of overall sales performance.
- Business strategy should focus on improving customer retention and engagement within the Consumer segment.

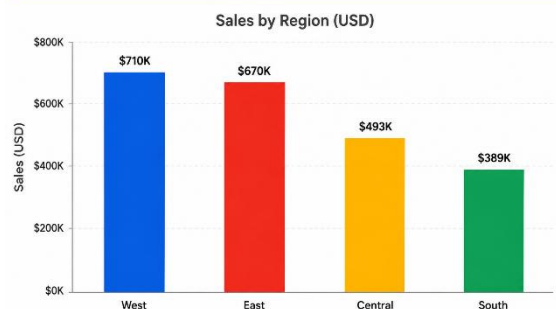
2. CUSTOMER SEGMENT ANALYSIS



3.Regional Performance

- West region generated the highest sales revenue at approximately **\$710K**.
- East region ranked second with nearly **\$670K** in sales.
- Central region contributed around **\$493K** in total revenue.
- South region recorded the lowest sales performance at approximately **\$389K**.
- Regional analysis indicates stronger customer demand and market performance in the West and East regions.

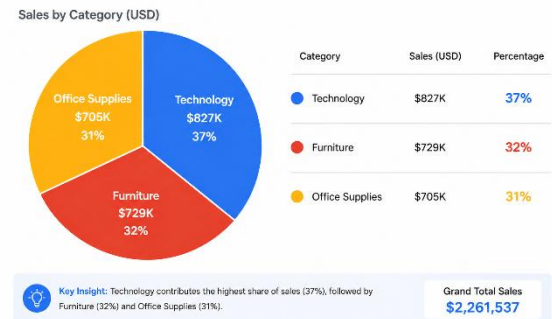
3. REGIONAL PERFORMANCE



4. Category Analysis

- Technology generated the highest sales contribution at approximately **37% (\$827K)**.
- Furniture contributed around **32% (\$729K)** of total revenue.
- Office Supplies accounted for nearly **31% (\$705K)** in sales.
- Technology remained the top-performing product category overall.

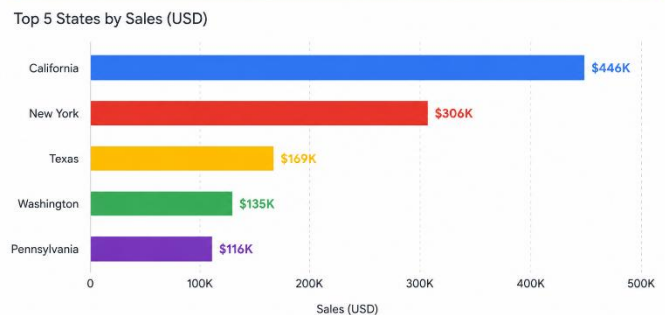
4. SALES BY CATEGORY



5. Geographic Analysis

- California generated the highest sales revenue, reaching approximately **\$446K**, making it the strongest-performing state.
- New York ranked second with nearly **\$306K** in total sales contribution.
- Texas, Washington, and Pennsylvania also showed strong market performance among the top-performing states.
- The results indicate that sales performance is highly concentrated in a few major geographic markets.

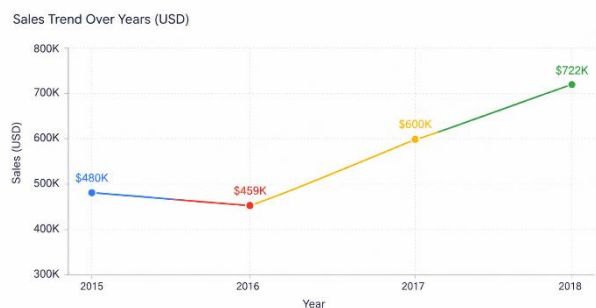
5. GRAPH SALES BY STATE



6. Sales Trends

- The business showed consistent year-over-year sales growth from **2015 to 2018**.
- Total sales increased from approximately **\$480K in 2015 to \$722K in 2018**.
- A slight decline was observed in **2016 (\$459K)** before strong recovery growth in later years.
- The highest yearly revenue was recorded in **2018**, indicating improved business performance and customer demand.
- Monthly analysis also showed that **November and December** generated peak sales, while **January and February** recorded comparatively lower sales performance.

6. YEARLY SALES TREND (2015–2018)



7.Shipping Analysis

- Standard Class generated the highest sales at approximately **\$1.34M**.
- Second Class contributed around **\$450K** in total sales.
- First Class recorded nearly **\$346K** in revenue contribution.
- Same Day shipping showed the lowest sales performance at approximately **\$125K**.
- The results indicate that customers mostly preferred cost-effective standard shipping methods.

7. SHIPPING MODE ANALYSIS



8. Recommendations (ACT Phase)

Based on the analysis, the following recommendations are suggested:

1. Increase Marketing in High-Performing Regions

Focus more investment on California and New York.

2. Strengthen Consumer Retention

Develop loyalty programs for high-value consumer customers.

3. Expand Technology Products

Increase inventory and promotions for Technology products.

4. Launch Seasonal Campaigns

Capitalize on strong Q4 demand patterns.

5. Improve South Region Performance

Investigate low-performing areas and improve market penetration.

6. Address Low Q1 Sales

Create promotional campaigns during January and February.

9. Future Improvements

Future versions of this project may include:

- Power BI dashboard development
 - Python-based exploratory data analysis
 - Sales forecasting models
 - Customer segmentation using machine learning
 - Streamlit web application deployment
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10. Conclusion

This project successfully analyzed e-commerce sales data using Microsoft Excel and SQL to identify meaningful business insights and performance trends.

The analysis highlighted strong-performing regions, top customer segments, leading product categories, and important seasonal sales patterns. Interactive dashboard visualizations further improved the understanding of business performance across different dimensions.

Overall, the project demonstrated how data analytics techniques can support data-driven business decision-making and improve strategic planning.

End of Report

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